

Elevate Labs — SQL Developer Internship

Task 2: Data Insertion and Handling Nulls

Submitted by: **K. Ameesha** | Key Concepts: DML Operations & Null Constraints

Part 1: Core SQL Concepts & Explanations

1. Difference between NULL and 0?

NULL indicates an unassigned, non-existent, or unknown value in a database column. It is not equivalent to zero or a blank string. Conversely, 0 is a defined numeric integer value. For instance, a student score of 0 means the exam was taken and scored zero, whereas NULL means the student was absent or the grade has not been entered.

2. What is a default constraint?

A DEFAULT constraint automatically supplies a predefined constant value to a column if no explicit value is specified during an INSERT query. It helps prevent unintended missing values and ensures data standardization across records.

3. How does IS NULL work?

Because SQL uses Three-Valued Logic (TRUE, FALSE, UNKNOWN), comparing a value to NULL using conventional relational operators (like = or !=) always results in UNKNOWN. The IS NULL and IS NOT NULL expressions specifically evaluate whether a database value is present or absent.

4. How do you update multiple rows?

Multiple rows can be updated at once by executing an UPDATE statement with a conditional WHERE clause that matches multiple records (or by omitting the WHERE clause to update the entire table).

5. Can we insert partial values?

Yes, partial insertion is allowed by explicitly listing the desired destination columns in the INSERT INTO statement. Any unlisted columns will be assigned their specified DEFAULT value or NULL if no default exists.

6. What happens if a NOT NULL field is left empty?

Leaving a NOT NULL field unassigned during an insertion (without a default fallback) causes the database system to throw a constraint error, instantly failing and rolling back the operation.

7. How do you rollback a deletion?

A deletion can be reverted if performed inside an active database transaction. By initiating the block with BEGIN TRANSACTION;, you can issue a ROLLBACK; statement to restore deleted rows before issuing a COMMIT;.

8. Can we insert values into specific columns only?

Yes, by specifying the target column names inside parentheses immediately following the table name, e.g.,
`INSERT INTO Table (ColA, ColB) VALUES (ValA, ValB);`.

9. How to insert values using SELECT?

Using the `INSERT INTO TargetTable (Cols...) SELECT Cols... FROM SourceTable` syntax, which copies records from a query result straight into the designated table.

10. What is ON DELETE CASCADE?

`ON DELETE CASCADE` is an option on foreign key constraints. When a record in a parent table is removed, all matching foreign key rows in related child tables are deleted automatically by the database engine.

Part 2: SQL Script & Practical Queries

Below is the fully functional SQL script demonstrating table setup, insert variations, null management, batch updates, and transactional deletes.

SQL IMPLEMENTATION SCRIPT (TASK2_AMEESHA_SUBMISSION.SQL)

```
-- =====
-- Elevate Labs SQL Developer Internship - Task 2
-- Student Name: K. Ameesha
-- Topic: Data Insertion, Handling Nulls, Updates, and Deletions
-- Compatible with: DB Fiddle / SQLite / MySQL
-- =====

-- 1. Table Setup
DROP TABLE IF EXISTS student_records;

CREATE TABLE student_records (
    student_id INTEGER PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    course VARCHAR(50) DEFAULT 'Computer Science',
    email VARCHAR(100) UNIQUE,
    gpa DECIMAL(3,2),
    enrollment_date DATE DEFAULT CURRENT_DATE
);

-- 2. Insert Complete Records
INSERT INTO student_records (student_id, full_name, course, email, gpa, enrollment_date)
VALUES
(101, 'K. Ameesha', 'Data Science', 'ameesha.k@example.com', 3.85, '2026-02-01'),
(102, 'Rohan Kumar', 'Information Tech', 'rohan.k@example.com', 3.40, '2026-02-02');

-- 3. Partial Value Insertions (Triggering DEFAULT and NULL)
INSERT INTO student_records (student_id, full_name, email)
VALUES
(103, 'Sneha Patel', 'sneha.p@example.com');
-- Result: course defaults to 'Computer Science', gpa becomes NULL

INSERT INTO student_records (student_id, full_name, course, gpa)
VALUES
(104, 'David Miller', 'Cyber Security', NULL);
-- Result: email becomes NULL, gpa is explicitly set to NULL

-- 4. Population via SELECT Query
```

```
CREATE TABLE temp_applicants (  
    app_id INT,  
    applicant_name VARCHAR(100),  
    contact_email VARCHAR(100),  
    score DECIMAL(3,2)  
);  
  
INSERT INTO temp_applicants VALUES  
(105, 'Ananya Ray', 'ananya@example.com', 3.90),  
(106, 'John Doe', 'johndoe@example.com', 0.00);  
  
INSERT INTO student_records (student_id, full_name, email, gpa)  
SELECT app_id, applicant_name, contact_email, score FROM temp_applicants;  
  
-- 5. Querying NULL Values  
SELECT * FROM student_records WHERE gpa IS NULL;  
  
-- 6. Batch Updating Missing Values  
UPDATE student_records  
SET gpa = 3.00  
WHERE gpa IS NULL;  
  
UPDATE student_records  
SET course = 'Advanced Data Science'  
WHERE course = 'Data Science';  
  
-- 7. Deletion with Transaction Control  
BEGIN TRANSACTION;  
  
DELETE FROM student_records  
WHERE gpa = 0.00;  
  
COMMIT;  
  
-- Final Result Check  
SELECT * FROM student_records;
```